Range Expansion, Cross-Asset Comparable-Vol Cohort (a05-volatility)

Strategy as submitted

Everything below is what the analyst submitted before the trading window ran. A result cannot be checked against a strategy that is not shown alongside it.

Discipline

volatility

Thesis

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Round 1's mechanism was post-shock drift: a true-range day well above its own ATR(14) that closes in t
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Universe

TLT, DBC, EWJ, XLU, XLP, INDA

Entry rule

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high - low > atr(14) * 1.2 and close > open
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Exit rule

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close < sma(10)
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Brackets and sizing

Stop loss	2.50%
Take profit	4.00%
Max hold days	20
Max positions	4
Position size (fraction of equity)	20.00%
Designed through	2026-02-13

Trading window

Designed through 2026-02-13. Traded 2023-08-21 through 2026-08-18, 751 sessions.

This run's trading window starts on or before the design cutoff, so the analyst designed this strategy with sight of part of the data it just traded. The portion of the run before 2026-02-13 is in-sample and does not measure out-of-sample performance -- only sessions after the design cutoff do.

Results

Total return	-2.23%
Max drawdown	-9.80%
Sharpe	-0.20

Sortino	-0.15
Volatility	0.04
Exposure	60.59%
Win rate	40.62%
Profit factor	0.92
Trade count	160

Reward:risk, planned versus realised

Planned is the ratio the card declared before trading (take-profit over stop-loss). Realised is what actually happened -- mean winning return over mean losing return -- since most trades exit on the rule or the time stop, not on either bracket.

Planned reward:risk (card)	1.60
Realised reward:risk	1.35

Benchmark comparison

Strategy total return	-2.23%
Buy-and-hold benchmark return, same sessions	75.40%
Excess over benchmark	-77.63%

Trades

Showing the first 50 of 160 trades.

Symbol	Entry	Exit	Qty	Entry $	Exit $	PnL	Return	Reason
TLT	2023-08-30	2023-09-05	20	$96.21	$94.23	$-39.66	-2.06%	rule
EWJ	2023-08-30	2023-09-21	32	$61.01	$61.78	$24.58	1.26%	rule
INDA	2023-08-30	2023-09-21	45	$44.17	$44.50	$14.79	0.74%	rule
TLT	2023-09-29	2023-10-02	22	$89.31	$87.78	$-33.72	-1.72%	rule
XLU	2023-10-04	2023-10-05	70	$28.58	$28.30	$-19.67	-0.98%	rule
TLT	2023-10-09	2023-10-10	23	$85.54	$85.81	$6.15	0.31%	rule
XLU	2023-10-09	2023-10-10	69	$28.59	$28.90	$21.32	1.08%	rule
INDA	2023-10-09	2023-10-10	44	$44.01	$44.45	$19.30	1.00%	rule
TLT	2023-10-10	2023-10-11	23	$85.89	$87.81	$44.10	2.23%	rule
TLT	2023-10-11	2023-10-13	23	$87.89	$87.56	$-7.65	-0.38%	rule
DBC	2023-10-20	2023-10-25	78	$25.28	$24.77	$-39.85	-2.02%	rule
TLT	2023-10-24	2023-10-25	23	$84.48	$83.98	$-11.56	-0.59%	rule
DBC	2023-10-26	2023-10-27	80	$24.67	$24.81	$11.13	0.56%	rule
XLU	2023-11-02	2023-11-10	66	$30.27	$30.12	$-9.97	-0.50%	rule
DBC	2023-11-24	2023-11-27	82	$24.12	$23.93	$-15.65	-0.79%	rule

Symbol	Entry	Exit	Qty	Entry $	Exit $	PnL	Return	Reason
EWJ	2023-12-04	2023-12-05	31	$62.35	$62.04	$-9.67	-0.50%	rule
XLP	2023-12-01	2023-12-07	28	$70.65	$70.46	$-5.38	-0.27%	rule
TLT	2023-12-04	2023-12-14	21	$92.54	$97.61	$106.41	5.48%	target
INDA	2023-12-01	2023-12-15	43	$46.15	$48.62	$106.14	5.35%	target
XLU	2023-12-14	2023-12-18	60	$33.20	$31.76	$-86.46	-4.34%	stop
EWJ	2023-12-08	2023-12-19	31	$62.51	$63.31	$24.74	1.28%	rule
TLT	2023-12-14	2024-01-02	20	$97.71	$98.18	$9.34	0.48%	rule
TLT	2024-01-04	2024-01-05	20	$97.47	$96.45	$-20.46	-1.05%	rule
XLU	2024-01-03	2024-01-12	62	$32.13	$31.67	$-28.58	-1.43%	rule
XLP	2024-01-03	2024-01-17	27	$73.01	$72.19	$-22.20	-1.13%	rule
EWJ	2024-01-09	2024-01-23	31	$64.07	$65.87	$55.74	2.81%	target
TLT	2024-02-02	2024-02-06	20	$96.32	$94.19	$-42.66	-2.21%	rule
XLU	2024-02-02	2024-02-06	63	$30.97	$30.07	$-56.76	-2.91%	stop
XLP	2024-01-31	2024-02-12	27	$73.48	$73.15	$-8.97	-0.45%	rule
TLT	2024-03-04	2024-03-13	21	$93.75	$94.45	$14.64	0.74%	rule
XLP	2024-02-23	2024-03-19	26	$74.63	$75.52	$23.08	1.19%	rule
XLU	2024-03-05	2024-03-19	63	$31.49	$31.67	$11.27	0.57%	rule
DBC	2024-03-12	2024-03-25	89	$22.30	$22.79	$43.54	2.19%	rule
XLU	2024-03-28	2024-04-11	61	$32.61	$32.58	$-1.90	-0.10%	rule
DBC	2024-04-05	2024-04-18	84	$23.75	$23.36	$-32.83	-1.65%	rule
DBC	2024-04-24	2024-04-25	84	$23.50	$23.48	$-1.75	-0.09%	rule
XLP	2024-04-22	2024-05-02	26	$74.63	$75.23	$15.54	0.80%	rule
XLU	2024-04-25	2024-05-08	59	$33.31	$34.68	$80.76	4.11%	target
XLP	2024-05-03	2024-05-24	26	$75.72	$77.13	$36.60	1.86%	rule
XLU	2024-05-10	2024-05-24	56	$35.89	$35.47	$-23.58	-1.17%	rule
XLU	2024-06-03	2024-06-06	55	$36.32	$35.63	$-38.01	-1.90%	rule
XLU	2024-06-11	2024-06-12	56	$35.14	$35.63	$27.38	1.39%	rule
XLP	2024-06-03	2024-06-13	25	$77.13	$76.64	$-12.31	-0.64%	rule
XLU	2024-06-25	2024-06-26	57	$34.82	$34.39	$-24.57	-1.24%	rule
XLP	2024-06-18	2024-06-27	25	$77.60	$77.19	$-10.31	-0.53%	rule
XLP	2024-07-08	2024-07-25	25	$77.48	$78.02	$13.44	0.69%	rule
DBC	2024-07-26	2024-07-29	89	$22.38	$22.28	$-8.97	-0.45%	rule
EWJ	2024-07-26	2024-07-29	29	$68.34	$68.87	$15.31	0.77%	rule
XLU	2024-07-12	2024-08-02	56	$35.51	$37.50	$111.37	5.60%	target
XLU	2024-08-02	2024-08-06	53	$37.54	$36.25	$-68.43	-3.44%	stop

Rejected orders

Orders the mock broker refused. A strategy whose orders were mostly rejected did not run as written, and omitting this list would describe a different strategy.

2 rejections.

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insufficient_cash
insufficient_cash
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Statistical significance

p-value (two-sided, binomial on the win/loss record): 0.02

q-value, corrected via Benjamini-Hochberg across the 24 strategies tried in this tournament: 0.04

Limitations

• 751 sessions is one market regime, not a sample of regimes. A strategy that worked here has not been shown to work in a rising-rate, high-volatility, or sideways regime -- only in whatever this one was.

• The traded universe was chosen because the user already trades these names. That is a selection effect, not a random or representative sample of the market, and it biases every result in this report in a direction that cannot be corrected after the fact.

• Fills are simulated: orders execute at the next session's open with deterministic adverse slippage applied by the mock broker. These are not real fills -- a live broker would fill some orders worse, and might reject orders this simulation accepted.

• This is a simulation of a trading strategy, not investment advice, and no figure in this report should be treated as a recommendation to trade.